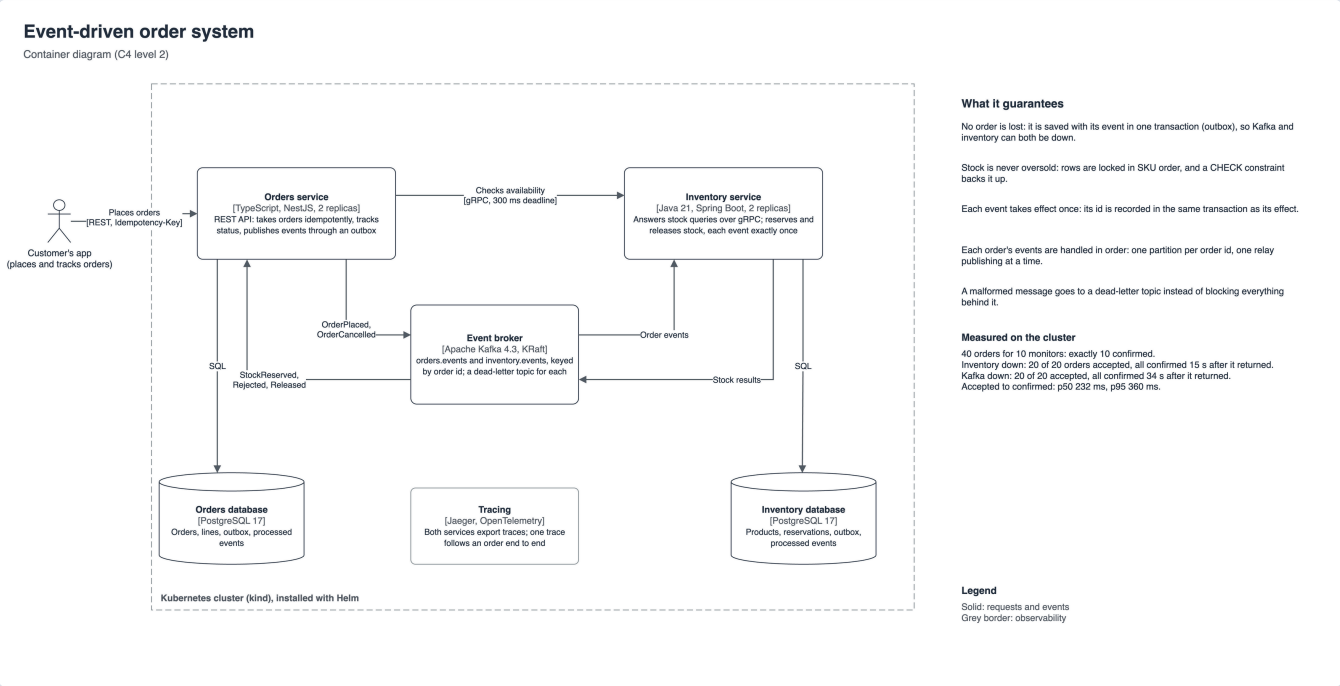
# Event-driven order system

Walkthrough in five parts: what each piece does, why it was built that way, and how to talk about it in an interview.

Repo: github.com/gumagatsby/event-driven-orders

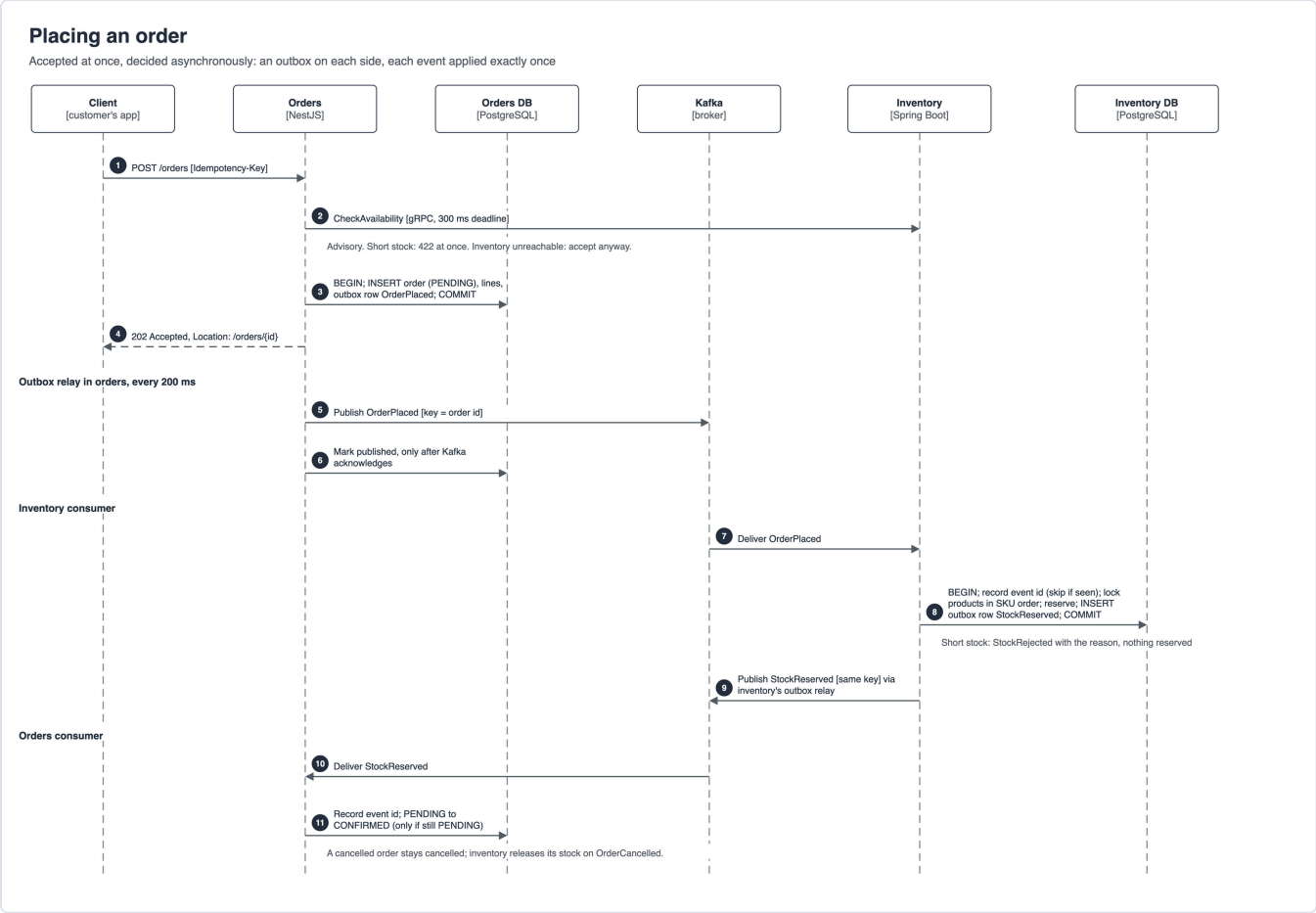



**The one-sentence pitch.** I built two microservices, one in TypeScript and one in Java, that take and fulfil orders over Kafka and gRPC on Kubernetes, designed so that an order is never lost even with Kafka or the inventory service down, stock is never oversold, and every event takes effect exactly once; then I proved it by taking each dependency down on a live cluster and checking every order came through.

## Part 1. What it is

| Piece | Technology | Job |
|---|---|---|
| Orders service | TypeScript, NestJS, node-postgres, kafkajs, grpc-js | REST API: takes orders idempotently, tracks status, publishes events, consumes stock results |
| Inventory service | Java 21, Spring Boot 4, Spring Kafka, Spring gRPC, Flyway | Answers stock queries over gRPC; reserves and releases stock from events |
| Event broker | Apache Kafka 4.3 (KRaft, no ZooKeeper) | `orders.events`, `inventory.events`, a dead-letter topic for each |
| Databases | PostgreSQL 17, one per service | Each service owns its data; a network policy enforces it |
| Platform | Kubernetes (kind, three nodes), Helm | One generic chart for both services, one for the infrastructure |
| Observability | OpenTelemetry, Jaeger, Prometheus metrics endpoints | One trace per order across both services and Kafka |

## Part 2. How an order flows




### Ask synchronously, decide asynchronously

Orders asks inventory over gRPC whether the stock exists, with a 300 ms deadline, so an impossible order is refused at once with `422`. That check is only advice. The decision is the reservation inventory makes when it consumes the `OrderPlaced` event. If inventory is down, the deadline expires and the order is accepted anyway; it is decided when inventory comes back.

### The outbox, on both sides

Neither service writes to Kafka from a request. It saves the event in an `outbox` table in the **same transaction** as the change, and a relay publishes it, marking it sent only after Kafka acknowledges. A crash cannot leave an order without its event, or an event for an order that was rolled back, and Kafka can be down without losing anything.

### Exactly-once effect on at-least-once delivery

Kafka can deliver a message twice. Each consumer records the event id in the same transaction as the event's effect and skips ids it has seen, so stock moves once however many times the event arrives.

### Order per order

Both topics are keyed by order id, so every event about one order is on one partition and is processed in order: placed before cancelled. Only one relay replica publishes at a time (a PostgreSQL advisory lock), which keeps that order on the way in.

## Part 3. Never oversold, never stuck

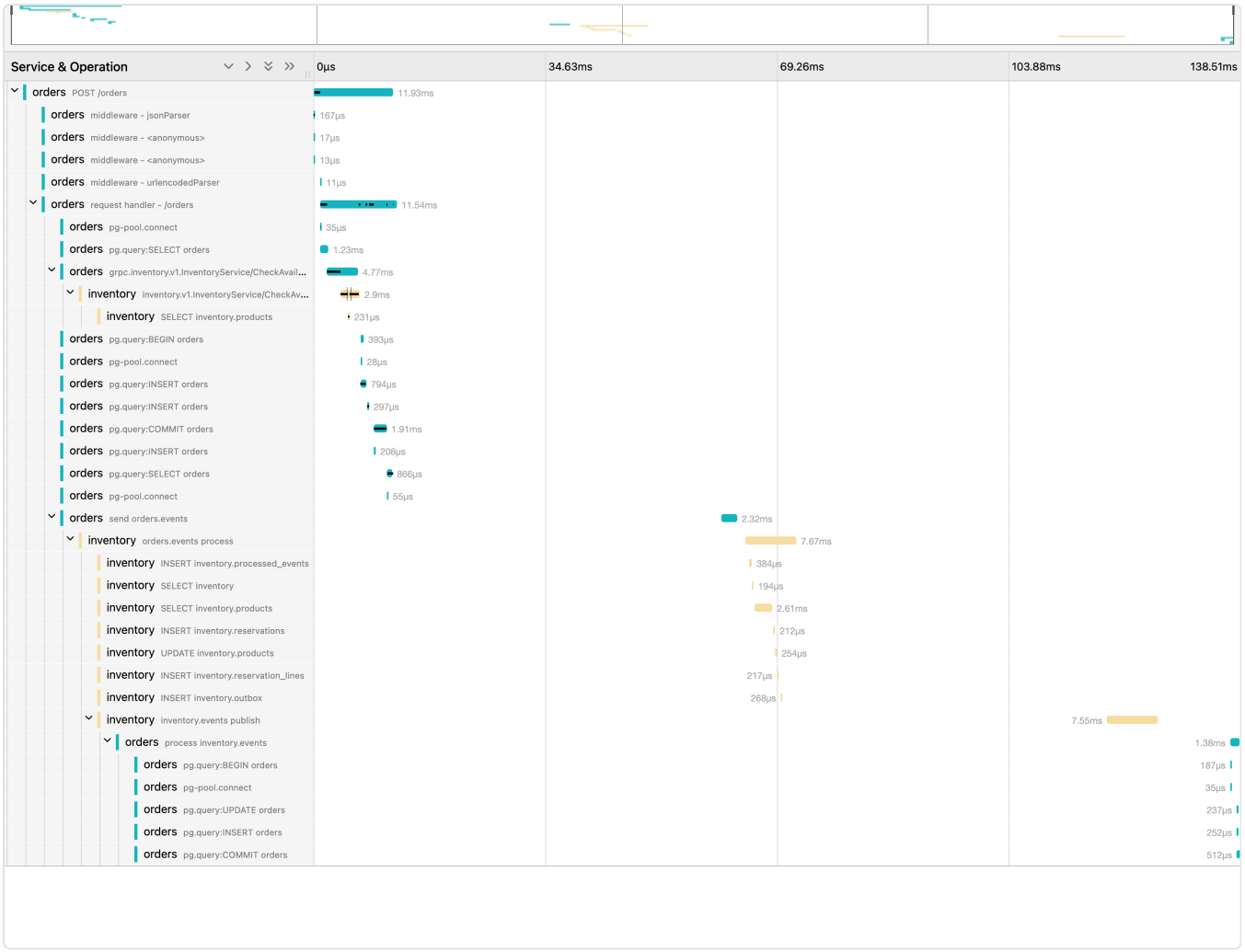
- **Locks in a fixed order:** inventory locks the product rows with `SELECT ... FOR UPDATE`, always sorted by SKU, so two orders for the same products cannot deadlock. `CHECK (reserved <= on_hand)` backs it up.
- **Poison messages:** a message that can never be parsed goes straight to a dead-letter topic, so it cannot block the partition. A temporary failure, such as the database being down, is retried with backoff for as long as it takes, because skipping it would lose a reservation.
- **Idempotent API:** `POST /orders` requires an `Idempotency-Key`, unique in the database. A retry returns the same order; the same key for a different order gets `409`.
- **Cancelled stays cancelled:** a stock result that arrives after a cancellation is ignored, because orders only moves an order out of `PENDING`.

## Part 4. Kubernetes

- **One chart for every service:** Deployment, Service, PodDisruptionBudget, NetworkPolicy and ServiceAccount, configured by a values file per service.
- **Locked-down pods:** non-root, read-only root filesystem, no Linux capabilities, no service-account token, seccomp on. Replicas spread across nodes; rolling updates never reduce capacity; a startup probe gives the JVM time to start.
- **Network policy, verified:** a pod without the orders label could not reach inventory's gRPC port, and inventory could not connect to the orders database.
- **gRPC balancing:** gRPC keeps one connection open, so a normal Service would pin all calls to one pod. A headless Service plus round-robin in the client spread checks 70 and 82 across two pods.
- **Secrets:** database passwords are generated by Helm on first install and kept on upgrade. None are in the repository.
- **A trap I hit:** Kafka never became ready. Its headless Service only published DNS for ready pods, and Kafka needed its own DNS name to become ready. `publishNotReadyAddresses: true` fixed it.

## Part 5. Proof

| Check on the cluster | Result |
|---|---|
| 40 orders race for 10 monitors | exactly 10 confirmed, 30 rejected, stock left 0 |
| Inventory scaled to zero, 20 orders placed | 20 of 20 accepted; all confirmed 15 s after it returned |
| Kafka scaled to zero, 20 orders placed | 20 of 20 accepted; all confirmed 34 s after it returned |
| 10 concurrent retries with one key | 1 order |
| Malformed message | dead-lettered; the next order still confirmed |
| Accepted to confirmed, 100 orders | p50 232 ms, p95 360 ms |




One order in Jaeger, 35 spans: the request (12 ms, including the gRPC check), the outbox relay sending to Kafka, inventory reserving stock, inventory's relay replying, and orders confirming, 138 ms end to end. The trace survives the outbox because each outbox row stores the trace context and the relay restores it.

**What the measurements caught.** After the first run, the traces were missing: Jaeger keeps traces in memory with no limit by default, the run pushed it past its memory limit, and Kubernetes killed it. I capped it at 20,000 traces. Then the first CI run failed one check: the catalogue, requested seconds after deployment, returned 503, because the orders service opened its gRPC connection on the first request and that took longer than the 300 ms deadline. It now connects at start-up.

---

## Interview questions

**"Why not just call inventory synchronously to reserve stock?"**

Then every order depends on inventory being up at that moment, and a timeout leaves you not knowing whether stock was reserved. With events, the order is safely recorded first and the reservation happens when inventory can do it. I kept a synchronous call only for the advisory check, with a deadline, so it can never block an order.

**"What is the transactional outbox and why do you need it?"**

You cannot commit to a database and publish to Kafka atomically. If you commit and then publish, a crash in between loses the event; if you publish first, a rollback leaves a false event. Writing the event to a table in the same transaction, and publishing from that table, removes the gap.

**"Kafka can deliver twice. How is that safe?"**

Each consumer records the event id in the same transaction as the effect, and skips ids it has seen. Delivery is at least once; the effect is exactly once. A test delivers the same event twice and checks stock moves once.

**"How do you keep events for one order in order?"**

Key by order id, so they share a partition, and have one consumer thread per partition. On the producing side, only one relay replica publishes at a time.

**"What would you change for production?"**

Three Kafka brokers with replication factor 3 or a managed service, a schema registry for the events, mutual TLS between services, alerts on consumer lag and outbox depth, and change data capture instead of polling to cut the latency.

---

## Checks done (16 October 2026)

| Check | Result |
|---|---|
| Java tests (Testcontainers: real Kafka 4.3.1 and PostgreSQL 17) | 12 of 12 pass, including the 40-way race and the dead-letter test |
| TypeScript tests (real PostgreSQL) | 20 of 20 pass, including 20 identical requests racing |
| End to end on a fresh three-node kind cluster | 13 of 13 checks pass, including both outage drills |
| Network policy probes | blocked and allowed paths as designed |
| Helm lint | both charts pass |
| GitHub Actions CI | all 4 jobs green; 13 of 13 end-to-end checks on a fresh cluster in CI |